

Quentin Goestchel

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I am currently a postdoctoral researcher, working on automatic detection and localization of marine mammals using Distributed Acoustic Sensing. I am interested in the analysis of acoustic data and applications to passive acoustic monitoring. My background is in acoustics, signal processing, and numerical modeling of sound propagation. I am deeply motivated by the development of open-source software that contributes to biodiversity conservation.

Education

- **Joint Research Unit in Environmental Acoustics (UMRAE)** **Strasbourg, France**
PhD degree in acoustics. Advisors: Gwenaél Guillaume, David Ecotièrre, Benoit Gauvreau *Oct, 2023*
Acoustic propagation in forest environments. Numerical study for environmental applications:
 - Theoretical study on the Transmission Line Matrix Method for modeling long-range forest scenarios.
 - Updating, improving and maintaining a code architecture in Python and OpenCL (C99).
 - Collaboration with bioacousticians (MNHN) for in-situ data.
- **Sorbonne Université** **Paris, France**
Master's degree in physical acoustics, Joint with ENSPS *Oct, 2020*
Graduated with highest honors.
- **Ecole Normale Supérieure Paris-Saclay (ENSPS)** **Cachan, France**
Master's degree of Ecole Normale supérieure Paris-Saclay *Oct, 2020*
Multidisciplinary 'Grande Ecole', specialization in engineering and research.
- **Lycée Eugène Livet** **Nantes, France**
Preparatory classes for engineering colleges *Jul, 2015*
Two-year undergraduate intensive course in Physics and Technology.
- **Lycée Aristide Briand** **St-Nazaire, France**
High School Diploma with a major in physics. Graduated with highest honors. *Sep, 2013*

Experience

- **University of Washington** **Seattle, WA, USA**
Postdoctoral researcher, PIs: Shima Abadi and William Wilcock *Jan, 2024 - Present*
 - Automatic detection and localization of marine mammals using DAS.
 - Contributing to the DAS4Whales open-source package.
- **Strasbourg University** **Strasbourg, France**
Computer science teacher (Lectures, tutorials and practical work) *Oct, 2022 - Feb, 2023*
- **Joint Research Unit in Environmental Acoustics (UMRAE)** **Strasbourg, France**
Trainee researcher *Mar, 2020 - Jul, 2020*
Numerical modeling of acoustic propagation over a forest floor using the TLM approach.
- **Eindhoven University of Technology (TU/e)** **Eindhoven, Netherlands**
Trainee researcher, supervisor: Maarten Hornikx *Sep, 2017 - Jun, 2018*
 - Applicability of the sound diffusion equation for acoustic simulations on 3D urban models.
 - Development of a finite volume method Matlab code for irregular tetrahedral meshes.
- **CERN, the European Organization for Nuclear Research** **Geneva, Switzerland**
Trainee engineer *Apr - Jun, 2017*
Modeling the noise impact of the LHC expansion (HL-LHC) with an engineering software.

Special skills

- **Languages:** French as mother-tongue, fluent in English (Cambridge Advanced C1), proficient in Spanish
- **Programming languages:** Python, C99 (OpenCL), Zsh, Bash, Matlab
- **Documents rendering languages:** $\text{\LaTeX}$, Markdown, reStructuredText
- **Softwares:** FreeCAD, Solidworks, Slurm Workload Manager, Git, Inkscape

National Scientific Communications

- **Technical days on acoustics and vibration** **Aix-en-Provence, France**
Virtual *Jun, 2023*
Oral presentation 'Propriétés du modèle TLM pour la propagation du son à l'extérieur : Effets de dispersion numérique'
- **Engineering Sciences Doctoral School Congress** **Vannes, France**
In-person *Jun, 2022*
Oral presentation 'Acoustic propagation in forest environments. Numerical study for environmental applications'
- **16th French Acoustics Society Congress** **Marseille, France**
In-person *Fev, 2022*
Oral presentation 'Stability analysis of TLM model for sound propagation in outdoor environment'
- **'Doctoriales' Planning, Mobility and Environment** **Le Croisic, France**
In-person *Oct, 2021*
Poster 'Acoustic propagation in forest environments. Numerical study for environmental applications'

International Scientific Communications

- **10th DCLDE Workshop** **Rotterdam, Netherlands**
In-person *June, 2024*
Oral presentation 'Comparison of Detection Techniques for Fin Whale Calls in a Distributed Acoustic Sensing Dataset: A Step Towards Automated Localization.'
- **186th Meeting of the Acoustical Society of America** **Ottawa, Canada**
In-person *May, 2024*
 - Oral presentation 'Automated Detection of Fin Whale Calls Recorded with Distributed Acoustic Sensing'
 - Oral presentation 'Sound propagation modeling in forests using the transmission line matrix Method: comparison with experimental in-situ measurements'
- **10th Forum Acousticum** **Torino, Italy**
In-person *Sep, 2023*
Oral presentation 'Transmission Line Matrix Method for sound propagation modelling In forests: comparison with in-situ measurements'
- **24th International Congress On Acoustics** **Gyeongju, South Korea**
Virtual *Oct, 2022*
Oral presentation 'Properties of the transmission line matrix model for outdoor sound propagation: Numerical dispersion effects'

Awards and Grants

- **Early Career Presenter Award in Computational Acoustics** **Ottawa, Canada**
Acoustical Society of America - Computational Acoustics Technical Committee *June, 2024*
- **UW eScience postdoctoral fellowship** **Seattle, WA, USA**
UW eScience institute *Jan, 2024*

Publications

Q. Goestchel, W. S. D. Wilcock, and S. Abadi. Enhancing fin whale vocalizations in distributed acoustic sensing data. May 2025.

L. Bouffaut, Q. Goestchel, R. A. Rørstadbotnen, A. Sladen, A. Hartog, and H. Klinck. Estimating sound pressure levels from distributed acoustic sensing data using 20 Hz fin whale calls. April 2025.

Quentin Goestchel. *Acoustic propagation in forest environments : time domain numerical methods toward bioacoustic applications*. phdthesis, Le Mans Université, October 2023.

Q. Goestchel, G. Guillaume, D. Ecotière, and B. Gauvreau. Transmission Line Matrix Model: Numerical Dispersion Effects on Simulated Specular Reflection [Accepted, to be published], March 2023.

Q. Goestchel, G. Guillaume, David Ecotiere, and B. Gauvreau. Properties of the Transmission Line Matrix Model for Outdoor Sound Propagation: Numerical Dispersion Effects. In *International Congress on Acoustics*, October 2022.

Q. Goestchel, G. Guillaume, D. Ecotière, and B. Gauvreau. Analysis of the Numerical Properties of the Transmission Line Matrix Model for Outdoor Sound Propagation. *Journal of Sound and Vibration*, page 116974, August 2022.

E. Walther and Q. Goestchel. The P.E.T. Comfort Index: Questioning the Model. *Building and Environment*, 137C:1–10, June 2018.

Extracurricular activities

- mountaineering, climbing, hiking, backcountry skiing, alpine skiing, sailing, bass guitar